Kiel Dang

Email: dang.v@northeastern.edu Phone: +1 647 782 4558 Medium LinkedIn Git-hub Portfolio

Data science practitioner with robust business acumen, boasting 2 professional working years of hands-on experience in data preprocessing and predictive modeling. Expert in SQL, Python, and R, with a track record of success in applying machine learning and statistical analysis to solve complex problems.

EDUCATION

Master - Analytics, Machine Intelligence **09.2021 – 07.2023**
Northeastern University, Toronto Campus

- GPA: 4.0/4.0 (97.89%) - Valedictorian.
- Top 20 finalists - VinUniversity Global Case Competition 2021 (VGCC): Electric Vehicle in the USA.

Bachelor - International Business Administration **08.2011 - 02.2015**
Foreign Trade University

- Merit-based Scholarship, 2013 and 2014.
- Talent Incubation Scholarship (Sponsored by Coca-Cola and Thanh Nien News for Top 5 students of the faculty), Nov 2013.
- Ambassador – Exchange participant in Indonesia, Jun to Aug 2013.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Alteryx, SAS, Excel VBA, SPSS, and Git.
Databases / Big Data: SQL Server, SQLite, Hadoop, and Spark.
Tools and Software: Jira, Bitbucket, GCP, Rstudio, Jupyter, R Studio, Tableau, Power BI, Alteryx, and SAS.
Specializations: Casual Framework, Branching Strategy, Data Science, Machine Learning, and MLOps.
Certificates: Alteryx Designer Core (May 2023).

PROFESSIONAL EXPERIENCE

Modeling and Advanced Analytics Co-op - *Definity Financial (Property & Casualty Insurance), Toronto.* **01.2023 - 05.2023**

Data Preparation

- Conducted 2 thorough dataset preparations for model development, including cleaning, feature engineering, and integration.

Model Development and Adaptation

- Developed and adapted 4 predictive models for underwriting, actuary, and claims handling, following pipeline and ensuring code reproducibility.
- Led model monitoring efforts, ensuring the performance and accuracy of predictive models for claims.

Data Analytics and Model Deployment

- Performed in-depth data analysis to uncover insights and patterns for actionable recommendations.
- Coordinated model development projects from initiation to completion and supported MLOps team to deploy models on GCP.

Hackathon winner: Form Autofill using Optical Character Recognition and Language Model (The first coop won the prize).
Tools: R, Python, Git, Jira, and Tableau to streamline workflows and pipelines, ensure version control, and facilitate collaboration.

Professor Assistant - *Northeastern University, Toronto* **09.2021 - 12.2021**

- Course: **Introduction to Statistics and Probability**
- Assisted the Professor with various tasks and duties related to technical support and mentoring for 30 students.

Senior Manager, Category Management - *Alibaba Group – Lazada E-commerce.* **09.2020 - 08.2021**

Data Analysis and Strategy Counselling

- Oversaw key e-commerce metrics of the top 80 sellers on platforms through dashboards, analyzing data to develop insightful analytics that captures problems.
- Optimized seller performances by proposing actionable insights derived from descriptive and diagnostic analysis.
- Performed ad-hoc analysis using hypothesis testing for business queries from business-as-usual activities.

Business Intelligence Projects

- Conducted A/B testing to evaluate the impact of key sales drivers (free shipping, vouchers, and hot deals) on revenue growth.
- Identified weighted factors influencing search exposure through data analysis, leveraging techniques such as text mining.

E-commerce Onboarding, Incubation, and Operation

- Managed, supervised, and trained 14 team members to ensure the smooth operation of Consumer-Packaged Goods category.

Tools: SQL Server, Power BI, Excel, and Ali Dashboard.

Manager, International Business Development - *VINAMILK, Asia Pacific* **08.2016 - 08.2020**

- Developed businesses of the company in international markets through exports and distributions by conducting market research, retail audit, and market penetration strategy.

HIGHLIGHTED PROJECTS

Resume and Job Matching (Python, NLP, Github, and Blog) **05.2023 – 07.2023**

- This application predicts the matching percentage of a candidate's resume to a job posting. It utilizes the Doc2Vec model and the Gale-Shapley algorithm.
- A nice visualization and notification tool was also launched to help candidates improve their CVs to get better chances.

Unsafe Excavation – Real-time Object Detection (Roboflow, NVIDIA Jetson, Github, and App) **04.2023 – 07.2023**

- Leveraged pre-trained models for object detection like YOLO, Mask R-CNN, and Roboflow platform to automate the workflow of real-time object detection.
- Achieved 91% mAP, 95% Precision, and 87% Recall.

Salary Prediction Application (Python, Streamlit, Docker, Azure, GCP, Github, and App) **10.2022 – 12.2022**

- The predictions were generated using an XGBoost model, while the web app was built on Streamlit framework. To ensure reproducibility, virtual environments were utilized on local hosts and contained by Docker. This app was deployed on GCP and Azure as well.
- A video guide on how to use the application is also available.

Time Series – Sale Forecasting (Python, Github, and Blog) **01.2022 – 03.2022**

- Evaluated stationarity, detected anomalies, decomposed data into components, tested white noises, and transformed data for modeling.
- Deployed SARIMAX to predict an 8-month sale forecast with the Mean absolute percentage of error of 17.06%.

Loan Default Classification (R, Github, and Blog) **05.2022 – 07.2022**

- Used many techniques including resampling, cost-sensitive logistic regression, and decision tree-based models with various methods of tuning such as loss matrix, adding weight or changing probability, XGBoost, and feature selection for Random Forest to tackle imbalanced class data.
- Achieved 93.65% of Sensitivity for performance metrics of minimizing Error type 2 in classification.
- Evaluated model performance by bad rate strategy, constructing a table with acceptance rates, corresponding cutoff values, and the percentage of bad loans that are accepted by management.

Inventory Management Database System for E-commerce Platform (SQLite, Github, and Blog) **10.2021 – 12.2021**

- Based on domain knowledge and the real operation of business to build a data system that recorded the flow of data and extracted data efficiently for business queries.
- Relational Database System design optimized to the 3rd Normal Form with a total of 15 entities for better data storage and maintaining data integrity that could facilitate data validation through SQL.

VOLUNTEER EXPERIENCE

Mobile Food Bank service, Red Cross Toronto, Canada **03.2023**

Sorted and classified food into portioned servings, ensuring equal access for all.

MLops World Conference & Expo 2022 Volunteer, Toronto **06.2022**

Played a crucial role as an organization support volunteer, ensuring smooth operations and efficient coordination at every step.

Course Registration and Orientation Day – Winter 2022 Volunteer, Northeastern University, Toronto **12.2021**

Coordinated with the school to help new students with course registration for the new academic semester and orientation day.